

A Detailed Literature Review

Visual Augmentation of Audio Semantics for Accessibility

Adam Gavriely

MSc Final Project — background reading, explained

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Abstract

This document is a self-contained, explanatory literature review for the project. Its goal is that reading it gives you the working understanding you would get from reading the primary papers: for each area and each key work it states the *problem*, the *core idea*, *how it works*, the *result/significance*, and *why it matters for us*. The project’s aim is to make the non-speech information in a video’s soundtrack — ambient and environmental sound — accessible to deaf and hard-of-hearing (DHH) viewers, by detecting the sounds a hearing viewer would notice, reasoning about which are *not* already conveyed by the picture, and rendering them as complementary visual augmentations shown alongside the video. We begin with the accessibility problem (the “why”), then work stage-by-stage through the technical building blocks (the “how”), and close by positioning the project against everything that already exists.

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1 Orientation: the problem and how the pieces fit

The project’s pipeline maps onto four mature research areas, which also organise this review: **audio event detection** (what sounds are present, and when?), **video understanding** (what is already visible?), **cross-modal reasoning** (which sounds are *not* conveyed by the picture, and how to depict them?), and **image/video generation** (show it). No existing system spans all four for accessibility. The review begins with the accessibility motivation (the “why”), then works through these building blocks (the “how”), and closes by positioning the project against what already exists.

2 Accessibility: subtitles, sign language, and the case for visual augmentation

This is the heart of the motivation, so it is treated first and in depth. The question is not merely “can we visualise sound?” but “*why* is a generated visual better than the accessibility tools DHH viewers already have?” Answering that requires understanding what subtitles and sign language do well, where they fail for non-speech sound, and what prior work has tried.

2.1 Subtitles and SDH: what they convey and where they break down

Plain subtitles transcribe *dialogue*. **SDH** (Subtitles for the Deaf and Hard-of-hearing) is the richer variant that is *expected* to represent non-dialogue audio too: sound effects, music, speaker identity, manner of speaking, and audience reaction. Industry style guides such as the **DCMP Captioning Key** [27] codify the conventions: sound effects in square brackets ([**glass shatters**]), music marked with note symbols, and — tellingly — the rule that the source of a sound effect should be named “*unless the source is clearly seen on screen*”, and that “if it can be inferred through the visuals, it is not necessary to caption it.”

That last rule is important for us: it means professional captioners already perform, by hand, exactly the *cross-modal gating* this project automates — describe a sound only when it is not already visible. Our Stage 5 is, in effect, an automation of an editorial principle the captioning profession already endorses.

Why SDH is nonetheless a poor channel for non-speech sound:

- **Low bandwidth / lost granularity.** A rich soundscape collapses to a terse label. A University of Sheffield study of captioned film found that in *Jaws*, captions told viewers there was “ominous music” but not that the music signified the shark *approaching* — so DHH viewers lost the suspense the score exists to create [29]. The label was present; the *meaning* was gone.
- **Emotional/atmospheric loss.** An EEG study by Gerber-Morón et al. [28] found that attending only to standard, compliant captions loses substantial emotional information, and that captions of non-verbal sounds tend to trigger extra *attentional* processing rather than an emotional response — i.e. the viewer reads “[thunder]” as a task, they do not *feel* the storm.
- **Divided attention.** Captions and the picture compete for one visual channel. Eye tracking shows caption users fixate the text a large majority of the time and watch the scene only peripherally [30]; the effort of reading causes fatigue and missed on-screen content (visual

gags, who is speaking). Systems like **LightSub** [31] exist specifically to reduce this eye-movement cost. This is a warning for us: a visual augmentation is *also* visual, so it must be designed not to worsen the very divided-attention problem it aims to relieve.

- **No spatial or temporal nuance.** Text cannot easily convey *where* a sound is (off-screen left? approaching?) or how it evolves over time.

2.2 Sign language: richer, but unavailable and not automatable

Sign language can carry far more of a soundtrack than bracketed text. Skilled interpreters convey music and sound through classifiers, mimed sound sources (an air-guitar, an instrument), sustained holds for long notes, rhythm, and **non-manual markers** — facial expression standing in for pitch and intensity — aiming to transmit the *feeling* of a sound, not just its name (as seen in theatrical and concert interpreting). So in principle sign language sets a high ceiling for expressive, affective conveyance of non-speech sound.

Its limits for this project’s setting are practical, not expressive:

- **Availability.** Interpreting is live, human and costly; the overwhelming majority of recorded/broadcast media has no interpreter, and where one exists the focus is speech (and sometimes music), rarely systematic ambient/environmental sound.
- **Automation is unsolved.** Automatic sign-language *generation* remains an open research problem, so it is not a deployable channel for ambient sound today.

The takeaway is that sign language shows what “good” looks like — rich, atmospheric, emotional — but cannot be delivered automatically or at scale for the non-speech layer.

2.3 Prior work on making sound accessible to DHH users

Several research threads have attacked pieces of this problem; none does what we propose, but each grounds it.

- **Sound-awareness systems (real environments).** Jain et al.’s **SoundWatch** [35] (a smartwatch that classifies and alerts on everyday sounds) and **HomeSound** [36] (in-home displays visualising household sounds, deployed with DHH families) establish, from field studies, *which* sounds DHH users want to know about — doorbells, appliance beeps, alarms, a baby crying, a dog, running water, sirens, a running vehicle. **ProtoSound** [37] lets users personalise a sound recogniser with a few examples. A large survey of DHH adults’ preferences for wearable/mobile sound-awareness technology [38] further found that users want the sound’s *identity* and *direction*, want filtering by *importance* (not every sound), and that preferences differ markedly across hearing profiles. Key lesson for us: these target *alerting/functional* sounds in the user’s own environment; *media* accessibility for *atmospheric/narrative* sound is a different, comparatively under-served need. But the sound taxonomy — and the “identity + direction + importance filtering” requirement — transfers.
- **Enriching captions (media).** A line of HCI work augments *text* with visual cues: visualising speech prosody and emotion in captions [32], visualising speech styles [33], and generatively adapting non-speech captions (**CapTune** [34]). These show the community sees the same gap we do — text alone is insufficient — but they stay within the caption/text paradigm and focus mostly on *speech* paralinguistics, not generated imagery for ambient sound.

2.4 Beyond subtitles and sign language: non-text visualizations and their measured effects

The user’s question “does a *visual* (not a caption, not an interpreter) actually help DHH viewers?” has been answered piecemeal across four literatures. Read together they form the strongest empirical support for this project.

(a) **Expressive (“affective”/“emotive”) captions — text made visual.** The earliest evidence that going beyond plain text helps comes from **emotive captioning** [39]: captions enhanced with graphics, colour and animation to carry sound effects, music and emotion. In the evaluation, eleven DHH viewers compared static and dynamic enhanced captions with conventional closed captions; *hard-of-hearing* viewers were significantly more positive and reported an enhanced viewing experience, while several *deaf* viewers disliked the style (seeing niche value, e.g. children’s programming). This deaf-vs-hard-of-hearing split is the field’s first clear demonstration that *DHH is not one audience* — a finding that recurs everywhere below. Modern follow-ups confirm the benefit with better designs: an emotion-visualising caption model outperformed conventional captions at conveying emotion and emphasis for 16 DHH participants at only a moderate legibility cost [32], and **Caption Royale** [40] mapped the design space of affective typography with 39 DHH participants across three studies, finding that users judge any visual enhancement first on **readability, minimal distraction, intuitiveness, and emotional clarity** — criteria that any visual channel we add must also satisfy.

(b) **Iconic and graphic sound representations in media.** Closer to our proposal, a thread replaces or supplements the *word* with a *picture*. An early pilot, **SoundSign** [41], rendered sound effects as icons with a compass showing direction and proximity. **Sound-word animation** [42] overlaid animated onomatopoeia (comic-style sound words) on videos; its user study found animated sound words “effectively and naturally visualise the dynamics of sound while clarifying the *position of the source*” and increase enjoyment and visual impact. However, formative work with Deaf participants warns that onomatopoeia is *phonetic*: “as a Deaf person, sometimes we might not understand the phonetic aspect of it” [34] — a person who has never heard “swish” cannot decode it. This is a direct argument for *depicting the source* (imagery), which requires no phonological knowledge, over sound-words. Most directly relevant, **Beyond Subtitles** [43] (Adobe Research, ASSETS 2022) interviewed 11 DHH viewers and built an authoring tool (on top of automatic sound event detection) for text *and graphic* captions of non-speech sounds — GIFs and animated stickers overlaid on video. Its findings: DHH viewers want *important* non-speech sounds represented; which representation is appropriate depends on the *sound type, the video genre, and the audience*; and there are real trade-offs between text (precise, but low-bandwidth) and graphics (evocative, but potentially ambiguous or distracting). Critical-design work at ASSETS 2023 [44] similarly probes richer renderings of non-speech information beyond bracketed text. Note, though, that in all of these the representation is *author-selected* and caption-anchored; none decides *automatically* which sounds need visualising given what the picture already shows.

(c) **First-person environments: AR, VR, and games.** Where sound carries actionable spatial information, visual substitution is the norm and is well measured. Jain et al. [47] designed head-mounted-display sound visualizations (arrows, pulses, varying directional granularity) with 24 DHH participants, who valued glanceable localization support. **SoundVizVR** [48] visualised loudness, duration, location and *type* (icons) of sound sources in VR: with 14 DHH participants, a full mini-map plus on-object indicators performed best, all icon variants let users locate sources, and in a follow-up VR game deployment most DHH players reported the indicators helped gameplay. Mirzaei et al. [49] compared visual (LED), haptic, and combined feedback for 3D sound localization: conditions *with the visual channel* were significantly faster than haptic alone — evidence that vision is the dominant usable substitute for spatial hearing. Industry has converged on the same conclusion at scale: Fortnite’s *Visualize Sound Effects* [50] renders a radial directional indicator with per-sound icons (footsteps, gunfire, chests), colour-coded threat and opacity-coded distance — a mainstream, shipping deployment of exactly the “icon + direction for sounds you cannot hear” idea, widely credited by DHH players. These systems prove that non-text sound visuals are *learnable, fast, and preferred* — but all operate in *first-person, live* settings (your environment, your game), not recorded media.

(d) **What DHH viewers say is still missing (field-wide surveys).** Two recent studies frame the gap our project targets. An analysis of non-speech audio captioning on YouTube [45] found most non-speech sounds are simply *not captioned at all*, with ambient/environmental audio the least covered category — the layer this project targets is the *most* neglected in practice. And a mixed-methods study combining a review of 36 papers with 168 DHH participants and 5 professional captioners [46] found DHH users want, beyond a text label: the sound’s *emotional tone*, its *spatial location*, and its *narrative significance* — and above all *customization* (“Choices? That’s the dream”).

Takeaways for this project. (1) The demand is real and the ambient layer is the least served: most non-speech sound goes uncaptioned [45], and what DHH viewers ask for — emotion, location, narrative significance [46] — is what a text label carries worst. (2) Non-text visuals demonstrably work for DHH users: icons and imagery convey source identity and location quickly [48, 47, 49], and animated visuals add dynamics and enjoyment [42, 39]. (3) Users judge any added visual first on readability and *minimal distraction* [40] — so *selection* (showing few, well-chosen augmentations) matters more than rendering; our cross-modal gate is precisely a selection mechanism. (4) DHH audiences are heterogeneous (deaf vs. hard-of-hearing reactions diverge [39]) and want control [46, 38] — augmentation should be a toggleable, complementary layer, which is why we composite *alongside* the video rather than altering the frame. (5) Phonetic devices (onomatopoeia) exclude some Deaf users [34]; *depicting the source* does not — supporting imagery over sound-words. (6) Every deployed non-text system is either first-person/live (AR/VR/games) or author-in-the-loop (graphic caption authoring); *automatic* visual augmentation of recorded media, gated by what the picture already shows, is the unoccupied slot this project fills.

2.5 The case for visual augmentation — a three-way comparison

Putting it together, the three channels have complementary strengths:

		Subtitles / SDH	Sign language	Visual augmentation (this project)
Speech		excellent	excellent	not the target (handled by captions)
Non-speech	ambient sound	terse label only	rich, if present	rich, generated depiction
Emotion	/ atmosphere	largely lost	strong	potentially strong (imagery)
Spatial	/ “where” info	none	possible (space/classifiers)	possible (directional cue)
Availability	/ scale	ubiquitous, cheap	rare, costly	automatable
Automatable	today	yes	no (generation unsolved)	yes (off-the-shelf models)
Added load	cognitive	competes with scene	separate focus point	must be designed carefully

Visual augmentation occupies the sweet spot: it is higher-bandwidth than an SDH label, automatable unlike interpreting, and it targets exactly the non-speech layer both other channels under-serve. It does not replace subtitles or sign language — it complements them, adding the missing “what does it look like / where is it / what mood does it set” that a label cannot

carry. Three design cautions follow directly from the literature above: (1) because it is visual, it must not worsen divided attention — users rank readability and minimal distraction above expressiveness [40] — which argues for *few, well-chosen* augmentations (aggressive gating) and a stable, peripheral placement; (2) atmospheric/emotional fidelity, not just correct labels, is what actually helps DHH viewers [46], which is why generation (imagery) beats classification (a word); (3) DHH audiences are heterogeneous and want control [39, 46], so the augmentation must remain an optional, complementary layer that leaves the original frame untouched.

Relevance to this project: This is the project’s justification. The novelty is not “visualise sound” (others visualise sound) but “*decide* which sounds are unconveyed by the picture and depict *those*, for media accessibility” — a combination no prior system provides.

3 Audio Event Detection: turning a soundtrack into labelled events

This is the engine that answers “what sounds are present, and when?”. The field is organised around one dataset and two model families.

3.1 AudioSet — the ontology everything is built on

AudioSet [1] (Gemmeke et al., Google, 2017) is a human-labelled dataset of ~ 2 million 10-second YouTube clips organised under an ontology of **527 sound classes** (speech, animals, vehicles, water, alarms, music, and so on). It is the de-facto vocabulary and training set for modern audio recognition. *Its strength and weakness for us:* broad coverage of everyday sounds (water, cat, dog, rain, sirens are all present), but a large fraction of the 527 are music/instrument classes, granularity is uneven, and it is a fixed, closed set.

3.2 PANNs — the practical workhorse (what we use)

PANNs [2] (“Pretrained Audio Neural Networks”, Kong et al., 2020) are CNNs trained on AudioSet, the best-known being **CNN14**. *How they work:* the audio is turned into a log-mel spectrogram (a time–frequency image), and a CNN scores each of the 527 classes. The clip-level model reports what sounds occur; the *DecisionLevelMax* variant produces *frame-level* outputs — a probability per class at each ~ 10 ms frame — giving time boundaries. PANNs reached a then-state-of-the-art mean average precision of **0.439** on AudioSet tagging. They are light, fast (sub-second on CPU for a short clip), and permissively usable. *Relevance to this project:* PANNs CNN14 (frame-level) is our implemented Stage-4 detector: it gives (label, start, end, confidence) events plus a timeline. It is the recommended first pass; its closed-set/granularity limits (e.g. it under-detects a regional air-raid siren) motivate an open-vocabulary companion.

3.3 AST and BEATs — transformer detectors (the quality path)

The **Audio Spectrogram Transformer (AST)** [3] (Gong et al., 2021) replaced the CNN with a pure transformer over spectrogram patches, capturing global context better. **BEATs** [4] (Chen et al., 2023) is the current practical state of the art: it is pre-trained with an *acoustic tokenizer* (a self-supervised scheme that learns discrete audio tokens, analogous to how language models are pre-trained), yielding strong general audio representations. BEATs is the baseline encoder in the DCASE sound-event-detection challenges, and appears again as the audio encoder inside audio captioners and even the TempoTokens video generator. *Relevance to this project:* BEATs is our planned Stage-4 quality upgrade from PANNs, and appears across several other systems we may use — a good encoder to standardise on.

4 Audio Captioning: describing a soundscape in words

Where tagging outputs labels, *automated audio captioning* (AAC) outputs a sentence: “a train approaches as birds chirp in the background.” This is a richer, compositional representation and a natural bridge to an image-generation prompt.

Datasets. **AudioCaps** [5] (~46 k AudioSet clips with human captions) and **Clotho** [6] are the standard benchmarks.

CLAP (Contrastive Language–Audio Pretraining) [7] is the workhorse underneath much of this: like CLIP for images, it learns a shared embedding space for audio and text, so you can measure how well a sound matches a text description — *including arbitrary descriptions it was never explicitly trained to classify*. This makes CLAP an **open-vocabulary** tool.

Modern captioners pair a strong audio encoder with a language decoder: **EnCLAP** [8] (neural audio codec + CLAP + a language model) and **SLAM-AAC** [9] (current state of the art on Clotho/AudioCaps, using paraphrase augmentation and CLAP-based re-ranking of candidate captions). *Relevance to this project:* Captioning gives us (a) a natural way to turn a detected sound into a generation prompt, (b) the required “audio captioning only” *baseline*, and (c) via CLAP, both an open-vocabulary detection option and an audio↔image relevance metric for evaluation.

5 Audio Understanding: large audio-language models (LALMs)

LALMs fold ASR, sound-event reasoning and captioning into one instructable model: an audio encoder + a connector + a large language model, so you can *ask* about a sound in natural language.

- **Qwen-Audio / Qwen2-Audio** [10] — strong general audio understanding; can be prompted both to transcribe speech and to describe non-speech sounds; outperformed prior methods across many audio benchmarks.
- **SALMONN** [11] — combines speech, audio-event and music encoders on a frozen LLM, giving general “hearing” abilities (ASR, translation, audio captioning) in one model.
- **Audio Flamingo 3** [12] — a fully open, long-context LALM supporting multi-turn, multi-audio dialogue.
- **MMAU** [13] — a benchmark for how well such models actually *reason* about audio, useful for justifying model choices.

Relevance to this project: A LALM could collapse our Stages 3–4 (ASR + event detection) into one component, and offers an *open-vocabulary* route (describe any sound in text). Trade-off: heavier and less individually inspectable than PANNs + Whisper as separate, swappable modules. Plan: keep the modular path as the default; treat a LALM (Qwen2-Audio) as an alternative backend and an ablation.

6 Vision-Language and Multimodal LLMs: understanding the picture

Two of our stages need to “see”: Stage 2 (what is already visible?) and Stage 7 (describe the augmented scene for evaluation).

- **Qwen2.5-VL** [14] — the leading open option: native dynamic-resolution vision, explicit *video* understanding with temporal grounding (it can point to *when* an event happens), released at 3B/7B/72B. The 7B fits a mid-range GPU; the 3B is a CPU/low-VRAM fallback.
- **InternVL3** [15] and the **LLaVA** [16] family are the main open alternatives; LLaVA (“visual instruction tuning”) popularised the connector- + -frozen-LLM recipe most VLMs now follow.

Relevance to this project: Qwen2.5-VL-7B is our candidate for Stage 2 (scene/visible-object description) and doubles as the Stage-7 *describer*. Because the same family spans understanding

and evaluation, we must use different instances/prompts for generation vs. judging so a model never grades its own output.

7 Sound-Source Localization: is the source on-screen or off?

This area is the technical key to our core “gap-aware” decision, and is easy to overlook. The question “is the thing making this sound actually visible in the frame?” has its own literature.

- **Sound-source localization via cross-modal alignment** [17] (Senocak et al., ICCV 2023) learns to point to *where* in an image a sound originates by aligning audio and visual features — and, by extension, whether such a location exists at all.
- **Object-aware localization** [18] grounds the sound in specific visible objects via audio-visual scene understanding.
- **Off-screen sound separation** [19] explicitly models the case where a sound has *no* on-screen source, estimating an on-screen-vs-off-screen probability and separating off-screen sound (sometimes using binaural cues for direction).

Relevance to this project: These give a *quantitative* “is the source visible?” signal — exactly our gate — without relying on an LLM’s judgement. Recommended design: use a localization signal to *ground* the LLM’s gating decision, and offer the ablation “LLM-only vs. LLM+localization”.

8 Multimodal Reasoning and its Failure Modes

Stage 5 asks a model to fuse scene, speech and audio events and decide what to depict. The literature warns this is where things break:

- Multimodal LLMs **hallucinate** — inventing content not in the input; “more thinking, less seeing” can amplify it. **Fork-Merge decoding** [20] mitigates this in audio-visual LLMs by reasoning over each modality separately before merging.
- Models suffer **cross-modal attention imbalance** — over-attending to one modality; “Pay More Attention To Audio” [21] documents and corrects this in audio-language models.

Relevance to this project: Direct implications for Stage 5: feed the LLM *structured* evidence (the visible-object list, the transcript, the event labels) rather than raw media; force structured JSON output with a required “reason” field; and forbid it from inventing sounds/objects not in its inputs.

9 Audio-to-Image Generation: showing a sound

Two philosophies, both relevant.

(1) **Direct audio-to-image.** **Sound2Scene** [22] (Sung-Bin et al., CVPR 2023) learns, from unlabelled video alone, to map audio into the latent space of a pretrained image generator and synthesise a matching scene directly from sound. **SonicDiffusion** [23] conditions a pre-trained diffusion model on audio. These are “sound in, picture out” with *no* language step and *no* reasoning about the existing video — which makes them the natural *direct-audio-to-image baseline* the proposal requires.

(2) **Text-conditioned generation.** **Stable Diffusion XL (SDXL)** [24] and **FLUX.1** [25] are the open state of the art in text-to-image. Our method sits here, but with the crucial twist that the *prompt is chosen by cross-modal reasoning* (Stage 5), not produced blindly from the audio. *Relevance to this project:* Generate augmentations with SDXL/FLUX from a Stage-5 prompt; use Sound2Scene as the “blind audio→image, no visual reasoning” baseline. The contrast — reasoned, gap-aware prompting vs. blind generation — is exactly what the evaluation isolates.

10 Audio-to-Video Generation: TempoTokens (in depth)

TempoTokens [26] (Yariv, Gat, Benaim, Wolf, Schwartz, Adi; AAAI 2024) is directly relevant and worth understanding fully.

Problem. Generate a short video that is aligned to an input audio clip both *globally* (the whole video matches the sound’s semantics) and *temporally* (each segment of audio maps to the corresponding segment of video) — for diverse, natural sounds, not just talking heads.

How it works. It reuses two frozen pretrained models and trains only a small bridge between them: (i) a frozen **BEATs** audio encoder produces embeddings; (ii) a lightweight **adapter** (“AudioMapper”, an MLP) maps those embeddings — using windows of varying sizes to capture local-to-global context — into *pseudo-text-tokens*, i.e. vectors in the same space the text-to-video model expects from a text prompt; (iii) a frozen **ModelScope** latent-diffusion text-to-video model turns those tokens into video. Because the audio is converted into the model’s “text token” language, the system can be conditioned on audio, text, or *both* (a first).

Evaluation and results. Trained/evaluated on **Landscape**, **AudioSet-Drums** and **VGG-Sound**. The authors introduce **AV-Align**, a metric that detects energy peaks in the audio and in the generated video and measures how well they line up in time — a way to quantify temporal synchrony. TempoTokens reports better audio alignment, visual quality and diversity than prior audio-to-video methods, plus a human study.

Limitations. It is bounded by the frozen T2V model’s capability; temporal conditioning is capped at 24 frames (short clips); and audio–visual discrepancies can arise (the sound may not fully determine the scene). It *generates/replaces* footage and never analyses an existing video — i.e. it does not do cross-modal gating. *Relevance to this project:* Three concrete uses: (1) the *direct audio-to-video baseline* (the video analogue of Sound2Scene, contrasting with our gap-aware method) — the primary, low-effort use; (2) an off-the-shelf motion-generation backend for our stretch goal; (3) its **AV-Align** metric for scoring temporal alignment of moving augmentations. It shares BEATs and VGGSound with us. It complements, not replaces, our contribution. (Co-authored by I. Schwartz.)

11 Datasets

- **AudioSet** [1] — 2 M clips, 527-class ontology; the training basis for PANNs/BEATs and our detection vocabulary.
- **AudioCaps** [5] / **Clotho** [6] — audio captioning benchmarks.
- **VGGSound** [51] — ~ 200 k 10-second YouTube clips, 309 classes, each with a *visible* sound source. Clean single-event ambient clips; its “source on screen” property can be inverted to build “source off-screen” test cases that stress our gate. Also used by TempoTokens.
- **UnAV-100** [52] — ~ 10.8 k untrimmed videos, 30 k+ audio-visual events, 100 classes, avg 2.8 (often overlapping) events per video; CC-BY 4.0. Ideal for *mixed* scenes with concurrent sounds.
- **MovieNet** [53] — 1,100 movies with rich annotations (42 k scene boundaries, character/place/action/style tags); best source for cinematic *acoustic-event* scenes with narrative context.
- **Landscape** / **AudioSet-Drums** — used by TempoTokens; narrower domains.

Relevance to this project: Our ~ 300 -clip benchmark samples VGGSound (clean ambient/off-screen cases), UnAV-100 (mixed/overlapping), and MovieNet (narrative acoustic events), across the three proposal scenarios.

12 Evaluation Methods

- **CLIP** [54] — shared image–text embedding; measures how well a generated image matches a text description (relevance).

- **CLAP** [7] — the audio analogue; measures audio $\leftrightarrow$ text (and, via a shared space, audio $\leftrightarrow$ image) relevance.
- **LLM-as-judge** — an independent LLM scores semantic consistency between the augmentation’s described content and a reference derived from the original multimodal input; the proposal’s automatic protocol (VLM-describe $\rightarrow$ LLM-judge). Must be validated against a small human-labelled subset and must not grade its own model’s output.
- **AV-Align** [26] — temporal alignment of audio and (generated) video via energy-peak matching; for evaluating *moving* augmentations.
- **Gating accuracy (ours)** — precision/recall of the “augment vs. stay silent” decision on a hand-labelled subset; the metric that most directly measures the core contribution.

13 Synthesis: where this project sits

Reading across the areas, the landscape is:

- **Accessibility** research has thoroughly documented that non-speech sound is poorly served by subtitles, that sign language is expressive but unavailable/unautomatable, and has measured — with DHH participants — that non-text sound visuals (icons, animated graphics, directional indicators) are learnable, fast and preferred in AR/VR/games and welcomed in media when author-selected. What it has *not* built is generated imagery for ambient sound in recorded media, selected *automatically* and gated by what is already visible.
- **Detection, captioning, LALMs, VLMs, localization** are all mature enough to assemble off-the-shelf; the pieces exist.
- **Generation** (audio-to-image, audio-to-video) exists but in a *blind* form — “sound in, picture out” — with no reasoning about the existing video. Those systems (Sound2Scene, Tempo-Tokens) are precisely our *baselines*.

The gap nobody fills is the combination: *detect non-speech sound, reason about the audio-visual gap (what is heard but not seen), and generate a complementary visual for accessibility*. That combination — task formulation, cross-modal gating, benchmark, and an evaluation protocol for it — is the contribution. Everything reviewed here is either a component we glue together or a baseline we measure against.

References

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